SYAM SUNDAR REDDY N

Data Scientist | GenAI & LLM Applications (RAG, Agents) | Python Backend Engineer
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SUMMARY

Data Scientist and Python backend engineer with ~4 years of software experience building production APIs and Generative AI systems, including Retrieval-Augmented Generation (RAG), LLM agents, and text-to-SQL, with measured evaluation. At Taskora, cut an enterprise synchronization workflow from 24 hours to 22 minutes (~65x) using parallel execution.

SKILLS

LLM & Agents: LLMs, Prompt Engineering, RAG, LangChain, LangGraph, AI Agents, MCP, Function Calling, Structured Outputs
Retrieval & Evaluation: Embeddings, Vector Databases, Semantic Search, Hybrid Search (BM25), Reranking, Metadata Filtering, RAG Evaluation
Backend: Python, FastAPI, REST APIs, SQL, PostgreSQL, SQLAlchemy, Alembic, Redis, JWT, OAuth 2.0, RBAC, Pytest
Infrastructure: Docker, CI/CD, AWS, Logging, Monitoring, Caching, System Design

WORK EXPERIENCE

Taskora Private Limited | Data Scientist — Mar 2025 – Present

- Re-engineered a large-scale enterprise synchronization workflow using parallel execution, reducing end-to-end processing from **24 hours to 22 minutes** (~65x faster, 98.5% reduction).
- Developed RAG-based AI applications using LLMs, embeddings, vector search, and contextual retrieval, enabling users to query enterprise information in natural language.
- Built scalable Python backend services and REST APIs for enterprise multi-tenant applications, supporting business-critical workflows, third-party integrations, and high-volume data processing.
- Worked with product and engineering teams to turn business requirements into data-driven and AI-powered features.

Innovious IT Services Pvt. Ltd. | Software Engineer — Sep 2022 – Jan 2025

- Developed and maintained Python backend services and REST APIs for enterprise applications, including database-driven workflows and third-party integrations.
- Designed database schemas and wrote SQL queries to support data-driven application features.
- Wrote unit tests, took part in code reviews, and helped resolve production issues while working in an agile team.

GENERATIVE AI PROJECTS

Enterprise Knowledge & RAG Platform **FastAPI | LangChain | PostgreSQL | Redis | Vector Database | Docker | AWS**

- Indexed 12,000+ document chunks through automated ingestion, chunking, metadata enrichment, embedding generation, and vector indexing pipelines.
- Improved Precision@5 from **72% to 89%** on a 200-question evaluation dataset using semantic retrieval, BM25 hybrid search, metadata filtering, and reranking.
- Reduced p95 response latency by 34% and token usage by 27% using Redis caching and context optimization; added JWT/RBAC, citations, RAG evaluation, Docker, and AWS CI/CD.

Agentic Customer Support Platform **LangGraph | MCP | FastAPI | PostgreSQL | Redis | LLMs**

- Achieved 88% end-to-end task completion across 120 customer-support scenarios covering knowledge retrieval, customer lookup, ticket handling, and escalation.
- Reached 93% tool-selection accuracy using structured outputs, deterministic routing, LangGraph state management, and tool validation.
- Reduced manual resolution steps by 42% using MCP-based tools, persistent state, retries, audit logging, and human-in-the-loop approval.

Enterprise SQL AI Agent **Python | LangGraph | FastAPI | PostgreSQL | Docker**

- Achieved 91% executable SQL accuracy across 150 benchmark business questions covering filters, joins, aggregation, grouping, and multi-table analysis.
- Improved successful query completion from 78% to 92% through schema-aware prompting, structured generation, validation, and automatic retry handling.
- Blocked 100% of destructive queries with read-only validation; median response time under 3 seconds; exposed authenticated FastAPI endpoints with Docker Compose and automated tests.

EDUCATION

Srinivasa Institute of Technology & Sciences | Bachelor of Technology — 2019 – 2022